
Guarded Block Transfers for Polytope Distance: Linear Convergence with Explicit Constants and Gap-Certified Screening for SVM Training

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Abstract

1 Hard-margin SVM training is the distance problem between two convex hulls, and
2 the Triangle Algorithm (TA) solves it with a certified duality gap from supporting
3 hyperplanes. We study the weight-transfer steps used by the enhanced TA and pro-
4 pose *guarded block transfers*: k MDM-style transfers per scan are aggregated into
5 a single exact line search, compared in $O(1)$ time against the single best transfer,
6 and replaced by an away step whenever the leading transfer is capacity-clipped.
7 Analysing the method on the Minkowski-difference polytope, where the objective
8 is 1-strongly convex and one pyramidal width governs the geometry, we prove
9 global linear convergence with explicit constants and without the combinatorial
10 swap-step bounds of pairwise Frank–Wolfe. We further prove that the TA’s cer-
11 tified gap equals the Frank–Wolfe gap over that polytope divided by the current
12 distance, which yields a screening activation theorem: gap-certified safe screening
13 removes every non-support point after an explicit $O\left(\frac{k}{\rho} \log \frac{1}{\tau}\right)$ iterations, τ being
14 the optimal-face score margin. Every inequality in the analysis has a numerical
15 check. On five LIBSVM benchmarks the solver matches the baselines’ solutions
16 to five digits wherever both converge, is faster than LIBSVM on the RBF hard
17 margin of gisette and $3\text{--}4\times$ faster than kernel SMO on kernel L_2 margins with
18 expensive kernel columns, and is one to two orders of magnitude slower on low-
19 dimensional kernel problems with nearly touching hulls and on dense-support L_2
20 soft margins, where the pyramidal width scales like $1/n$ and LIBLINEAR finishes
21 in seconds. We report both sides.

22 1 Introduction

23 Training a linear hard-margin SVM is equivalent to finding the closest points of the two classes’
24 convex hulls: the optimal separator is the perpendicular bisector of the witness pair, and the margin
25 is half the hull distance [6]. Kalantari’s Triangle Algorithm (TA) exploits this *distance duality*
26 directly [1], maintaining an iterate pair (p, q) and a certified interval $[LB, UB]$ on the distance from
27 parallel supporting hyperplanes, so it can stop at a provable relative accuracy ε . The enhanced TA of
28 [2] adds joint segment updates, dot-product caching, an anti-zig-zag heuristic and prioritised search,
29 and reports large speedups over SMO on synthetic data.

30 The workhorse step of such geometric solvers is the *weight transfer*: move mass from the worst
31 active vertex to the best candidate vertex, the Mitchell–Demyanov–Malozemov (MDM) step [4],
32 which is exactly the pairwise Frank–Wolfe (FW) step of [3] on the relevant polytope. Each step
33 costs an $O(n)$ scan and produces one transfer; we ask how much of the scan can be reused across
34 several transfers without weakening the guarantee.

35 Contributions.

- 36 1. **Guarded block transfers** (§3): pair the top- k receivers with the worst- k active donors,
 37 take one exact line search along the aggregated direction, and *guard* by an $O(1)$ comparison
 38 with the single MDM step; on capacity-clipped iterations, fall back to an away step. Cache
 39 updates for a whole block are one GEMV.
- 40 2. **Linear convergence with explicit constants** (Theorem 1): on the Minkowski-difference
 41 polytope $Z = \text{conv}(V) - \text{conv}(W)$ the guarded block algorithm inherits the pyramidal-
 42 width rate of pairwise FW up to a constant, and the away fallback replaces pairwise FW’s
 43 combinatorial swap-step bound with linear drop-step counting.
- 44 3. **The certified gap is the FW gap** (Lemma 3), which yields a *screening activation* theorem
 45 (Theorem 2): gap-certified safe screening provably removes all non-support points after an
 46 explicit number of iterations.
- 47 4. **A regime study on real data** (§5). The analysis predicts $O(n^2)$ -type cost when the optimal
 48 support is dense ($PW \sim 1/n$). Two batteries on five LIBSVM datasets show the boundary
 49 from both sides: the solver wins or ties on sparse-support high-dimensional cells and loses
 50 by orders of magnitude on dense-support L_2 soft margins and on near-touching kernel
 51 hulls.

52 Code, proofs, numerical checks and experiment scripts are in the supplementary repository
 53 (anonymised for review).

54 2 Setting

55 Let $V = \{v_1, \dots, v_n\}$, $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$. The polytope distance problem is

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} \frac{1}{2} \|p - q\|^2. \quad (1)$$

56 Writing $z = p - q$ and $Z := \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$, (1) is the minimum-norm-point
 57 problem $\min_{z \in Z} F(z)$, $F(z) = \frac{1}{2} \|z\|^2$, whose objective is 1-strongly convex and 1-smooth. Let z^*
 58 be the minimiser, $\delta^* = \|z^*\|$, $h(z) = F(z) - F(z^*)$, $M = \text{diam}(Z)$ and $\delta = \text{PW}(Z) > 0$ the
 59 pyramidal width [3].

60 The algorithm keeps simplex weights α on V , β on W , with $p = \sum \alpha_i v_i$, $q = \sum \beta_j w_j$; the induced
 61 product representation $\lambda_{ij} = \alpha_i \beta_j$ of z over the atoms $v_i - w_j$ has support $\text{supp } \alpha \times \text{supp } \beta$. With
 62 $g = -\nabla F(z) = q - p$, the *side scores* are $s_i^V = \langle g, v_i \rangle$ and $s_j^W = \langle p - q, w_j \rangle$; the implementation
 63 maintains them incrementally as differences of cached arrays Vp, Vq, Wp, Wq .

64 **Certificate.** For any iterate, $\text{UB} = \|z\|$ and $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\| = (\min_i \langle z, v_i \rangle -$
 65 $\max_j \langle z, w_j \rangle) / \text{UB}$ satisfy $\text{LB} \leq \delta^* \leq \text{UB}$; the TA stops when $\text{UB} - \text{LB} \leq \varepsilon \text{UB}$. The L_2 soft
 66 margin, ν -SVM and kernel variants reduce to (1) exactly (augmented coordinates giving $K + I/C$;
 67 reduced convex hulls; feature-space images), see Appendix B.

68 **Gaps.** $g^{\text{FW}} = \max_{u \in Z} \langle g, u \rangle - \langle g, z \rangle$, $g^{\text{A}} = \langle g, z \rangle - \min_{u \in \text{supp } \lambda} \langle g, u \rangle$, $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}}$.
 69 Because $\langle g, v_i - w_j \rangle = s_i^V - s_j^W$ is separable and the support is a product set, each gap splits into a
 70 V -side and a W -side term computable from the score arrays in $O(n + m)$ (Lemma 4), and a transfer
 71 $v_r - v_u$ on the V side is a pairwise direction of Z (Lemma 5).

72 3 Guarded block transfers

73 Algorithm 1 amortises one $O(n)$ scan over up to k transfers. Every candidate gain is available in
 74 $O(1)$ from cached quantities: with $S = \langle g, d \rangle$ and $D = \|d\|^2$, the block gain is $tS - t^2 D/2$ at
 75 $t = \min(S/D, 1)$, where $S = \sum_j \gamma_j \text{gap}_j$ and $D = \gamma^\top G \gamma$ for the $k' \times k'$ Gram matrix G of the
 76 pair directions, read off the cached columns; the MDM, away and toward gains are the standard one-
 77 dimensional expressions. The cache update for the whole block is a single matrix–vector product
 78 with the net per-index coefficients. Feasibility is immediate: donors lose at most $t\gamma_j \leq c_j$, receivers
 79 only gain, and the weight sum is conserved.

Algorithm 1 One iteration of the guarded block-transfer step (side V ; W symmetric)

```

1: top- $k$  receivers by  $s^V$ ; worst- $k$  donors in  $\text{supp } \alpha$ ; pair best-with-worst:  $(r_j, u_j)$ ,  $j \leq k'$ ; pair 1
   is the MDM pair  $(r^*, u^*)$ 
2:  $d_j \leftarrow v_{r_j} - v_{u_j}$ ;  $\text{gap}_j \leftarrow s_{r_j}^V - s_{u_j}^V$ ;  $\gamma_j \leftarrow \min(\text{gap}_j / \|d_j\|^2, c_j)$   $\triangleright c_j$ : donor capacity
3: if pair 1 uncapped ( $\gamma_1 = \text{gap}_1 / \|d_1\|^2$ ) then
4:    $\Delta_B \leftarrow$  gain of exact line search  $t \in [0, 1]$  along  $d = \sum_j \gamma_j d_j$ ;  $\Delta_1 \leftarrow$  gain of the MDM
   step
5:   take the step with the larger gain  $\triangleright$  guard,  $O(1)$ 
6: else
7:   if the away step on  $u^*$  is boundary-clipped then take it  $\triangleright$  drop step:  $\alpha_{u^*} \rightarrow 0$ 
8:   else take the best of {block, MDM, away( $u^*$ ), toward( $r^*$ )}
9:   end if
10: end if

```

Safe screening. With $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$ and $v_{\min} = \arg \min_i \langle z, v_i \rangle$, a zero-weight point with $\langle z, v_i \rangle - \langle z, v_{\min} \rangle > r \|v_i - v_{\min}\|$ is outside the optimal support and is removed (symmetrically on W with the max). Because the optimum's support survives, the reduced problem has the same optimum and all subsequent lower bounds remain valid.

4 Analysis

Full proofs are in Appendix A; we state the results and the ideas.

Lemma 1 (Block gain). *Let $b_j = \gamma_j \|d_j\|$. Then $S \geq \sum_j b_j^2$, $D \leq k' S$, and the realised block gain satisfies $\Delta_B \geq S / (2k') \geq \Delta_{\text{MDM}} / (2k')$.*

The proof is the triangle inequality on $\|d\|$ and Cauchy–Schwarz on $(\sum b_j)^2$; capacity clipping only helps. With the guard, every case-(a) iteration's gain is $\max(\Delta_B, \Delta_1) \geq \Delta_1$: the guarded step never makes less progress than pairwise FW.

Lemma 2 (Good-step progress). *Work on the side with the larger pairwise gap. On any non-drop iteration, $\text{gain} \geq \min(g^{\text{PW}^2} / (32M^2), g^{\text{PW}} / 8)$.*

Case (a) uses $g_{\text{side}}^{\text{PW}} \geq g^{\text{PW}} / 2$ and the exact line search bound; case (b) uses $g_{\text{side}}^{\text{PW}} = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$, so one of the toward step (always feasible with step ≤ 1) or the interior away step realises at least a quarter of g^{PW} .

Theorem 1 (Global linear convergence). *Run the guarded block-transfer algorithm with away fallback, one side per iteration, any $k \geq 1$, from any vertex pair (initial support $s_0 \leq 2$). Then every non-drop iteration satisfies $h_{t+1} \leq (1 - \rho)h_t$ with*

$$\rho = \frac{1}{16} (\delta / M)^2,$$

the number of drop steps is at most $k T_{\text{good}} + s_0$, and hence $h_T \leq h_0 (1 - \rho)^{(T-s_0)/(k+1)}$. The implementation, which steps both sides every iteration, satisfies the same statement with $\rho_{\text{impl}} = \frac{1}{64} (\delta / M)^2$ and exponent $(T - s_0) / (2k + 1)$ (Lemma 7).

The rate follows from Lemma 2 and the geometric strong-convexity bound $g^{\text{PW}} \geq \delta \sqrt{2h}$ (Lemma 6, from the pyramidal-width definition of [3] and 1-strong convexity), together with $g^{\text{PW}} \geq g^{\text{FW}} \geq h$ for the linear branch. The drop counting is the away-step argument: each drop removes one support index, non-drop steps add at most k , so drops are at most k times the good steps plus s_0 . There is no swap-step bound because case (b) never takes a capacity-clipped pairwise step: it takes an away step, which either contracts or drops.

Proposition 1 (Near-orthogonal speed-up). *If $|\langle d_i, d_j \rangle| \leq \eta \|d_i\| \|d_j\|$ for $i \neq j$, the interior block gain is at least $(\sum_j \Delta_j) / (1 + \eta(k' - 1))$, where Δ_j is pair j 's individual gain.*

As $\eta \rightarrow 0$ a single scan realises the sum of all k pairwise gains; in high dimension distinct support directions are nearly orthogonal, which is the mechanism behind the measured 4–30 $\times$ iteration reductions.

Table 1: Regime battery on real data: hard margins and kernels, five seeds, 600 s per solver, mean $\pm$ 95% CI. Kernel cells (RBF, $\gamma = 1/d$) use a class-balanced subsample of n_k training points. ETA and its baseline solve the same objective: LIN, linear hard margin (ETA with screening vs LIBSVM at $C=10^6$; our SMO in brackets); KHM, feature-space hard margin (K-ETA vs LIBSVM at $C=10^6$); KL2, kernel L_2 margin $K + I/C$, $C=1$ (K-ETA vs kernel SMO). conv = seeds in which ETA reached the certified gap; oracle = $O(nd)$ column (ETA) or row (SMO) evaluations, in thousands.

cell	dataset	n_k / d	ETA (s)	baseline (s)	conv	dist. ETA / base	acc. ETA / base	oracle
LIN	gisette	6000 / 5000	56.6 $\pm$ 5.9	51.4 $\pm$ 13.1 [405 $\pm$ 62]	5/5	3.6638 / 3.6640	.981 / .981	1.5 [11.1]
KHM	gisette	6000 / 5000	86.2 $\pm$ 14.8	104.6 $\pm$ 20.2	5/5	.05215 / .05215	.979 / .979	2.6
KHM	ijcnn1	9853 / 22	421 $\pm$ 149	1.3 $\pm$ 0.3	4/5	.00430 / .00430	.946 / .946	2.5
KHM	a9a	10000 / 123	> 600	26.4 $\pm$ 2.9	0/5	$\leq$.00009 / .0033	.690 / .752	5.9
KHM	covtype	10000 / 54	> 600	144 $\pm$ 7 (maxiter)	0/5	$\leq$.00053 / .00025	.704 / .712	7.2
KHM	w8a	6479 / 300	101 $\pm$ 159	—	5/5	hulls intersect	—	1.9
KL2	gisette	6000 / 5000	131 $\pm$ 34	420 $\pm$ 49	5/5	.06882 / .06882	.971 / .971	3.6 / 15.2
KL2	w8a	6479 / 300	11.5 $\pm$ 2.6	41.7 $\pm$ 10.0	5/5	.05682 / .05682	.975 / .975	3.6 / 20.2
KL2	a9a	10000 / 123	52.8 $\pm$ 71.0	57.7 $\pm$ 23.2	5/5	.03019 / .03019	.795 / .795	7.6 / 43.6
KL2	covtype	10000 / 54	26.0 $\pm$ 4.2	17.6 $\pm$ 0.5	5/5	.02609 / .02609	.792 / .792	9.1 / 56.2
KL2	ijcnn1	9853 / 22	9.2 $\pm$ 2.5	6.1 $\pm$ 0.7	5/5	.04790 / .04790	.959 / .959	4.6 / 26.1

113 **Lemma 3** (The certified gap is the FW gap). *At any full scan, $UB - LB = g^{\text{FW}}(z)/UB$, and*
114 *$g^{\text{FW}}(z) \leq 2h + R\sqrt{2h}$ with $R = \max_{u \in Z} \|u\|$.*

115 **Theorem 2** (Screening activation). *Assume $\delta^* > 0$, let F_V, F_W be the optimal faces, $\tau > 0$*
116 *the minimal nonzero score margin $\langle z^*, v_i \rangle - \min_j \langle z^*, v_j \rangle$ (and symmetrically on W), and D*
117 *the larger class diameter. A zero-weight point with margin τ_i is screened as soon as $r \leq$*
118 *$\tau_i/(3D)$; consequently, once $h_t \leq \tau^4 \delta^{*2}/(2592 R^4 D^4)$, which under Theorem 1 holds for all*
119 *$t \geq s_0 + \frac{k+1}{\rho} \log(2592 R^4 D^4 h_0/(\tau^4 \delta^{*2}))$, every zero-weight point outside $F_V \cup F_W$ is removed*
120 *permanently, and each later scan costs $O(|F| + |\text{supp}|)$ instead of $O(n + m)$.*

121 The τ -dependence is unavoidable: no rule that is safe for all instances consistent with a certificate of
122 gap $\sim \tau$ can distinguish a point of margin τ from a support point. Each of Lemma 1, Proposition 1,
123 Lemma 3, Lemma 7 and Theorem 2 is verified numerically on random and adversarial ensembles at
124 machine precision (Appendix D).

125 5 Experiments

126 **Setup.** All solvers are validated against an exact QP (SLSQP over the two simplices) on small
127 instances: distances agree to 10^{-9} or better for every variant. Stopping rule $\varepsilon = 10^{-3}$ on the cer-
128 tified relative gap throughout unless stated; baselines use matched tolerances on the same objective
129 (Appendix C). Synthetic instances follow the protocol of [2] (two Gaussian clouds, $n = 5,000$ per
130 class). Real data are the LIBSVM sets a9a, w8a, ijcnn1, gisette and covtype (100k subsample),
131 standardised, with official test sets where they exist, five seeds, and a 600 s budget per solver and
132 cell. All real-data runs share one dedicated 32-vCPU Xeon 2.6 GHz machine (122 GB, OpenBLAS,
133 one BLAS thread per solver, 26 cells concurrently), so times are comparable within a battery. Every
134 ETA-versus-baseline pair solves the *same* objective; a differently regularised solver appears only as
135 a labelled reference.

136 **Synthetic: steps, blocks, screening.** Away and pairwise steps converge at $\varepsilon = 10^{-6}$ where the
137 toward and midpoint steps of [2] do not, and block transfers cut iterations a further 4–30 $\times$ relative
138 to single MDM. Screening reduces 10,000 points to 36–1,054 survivors and wall-clock by 1.4–2.4 $\times$
139 at $\varepsilon = 10^{-5}$, and is neutral at 10^{-3} , where runs end before the certificate activates, as Theorem 2
140 predicts. On seven synthetic and small-real problem classes the optimised solver is fastest to a
141 certified solution on six (1.6–2.5 $\times$ ahead of LIBSVM, LIBLINEAR or NuSVC); the exception,
142 MNIST odd-vs-even, is a dense-support L_2 problem. Details and figures are in Appendix C.

143 **Real data, sparse-support regime (Table 1).** TA I first certifies linear separability: gisette is
144 separable and a9a, w8a, ijcnn1 and covtype are not (hulls intersect, certified in 0.2–42 s), a diagnostic
145 none of the baselines produces. On gisette ($d=5,000$; 1,233 of 6,000 points in the support) ETA ties
146 LIBSVM on the linear hard margin, both 7 $\times$ ahead of our SMO; K-ETA is 1.2 $\times$ faster than LIBSVM

Table 2: L_2 soft-margin battery (dense-support regime): five seeds (four for `ijcnn1`), 600 s per solver. ETA and our SMO time out on every cell; we report LIBLINEAR’s time, ETA’s certified relative gap at the cap (mean [min, max]) and test accuracies at the cap. The `gisette` and `covtype` cells and the fifth `ijcnn1` seed had not finished when this draft was prepared and are omitted.

dataset (n / d)	C	LIBLINEAR (s)	ETA gap @ 600 s	acc. ETA	acc. SMO	acc. LIBLINEAR
a9a (32,561 / 123)	0.1	3.7 ± 0.8	0.15 [0.10, 0.23]	.849	.845	.849
	1	6.8 ± 1.7	1.26 [1.25, 1.28]	.810	.780	.849
	10	2.3 ± 0.5	1.83 [1.80, 1.86]	.635	.763	.849
ijcnn1 (49,990 / 22)	0.1	0.4 ± 0.1	0.011 [0.007, 0.016]	.918	.918 [†]	.918
	1	0.7 ± 0.1	0.70 [0.61, 0.73]	.905	.917	.918
	10	0.4 ± 0.1	1.36 [1.34, 1.40]	.707	.860	.918
w8a (49,749 / 300)	0.1	22.7 ± 4.3	0.57 [0.50, 0.66]	.987	.985	.987
	1	20.6 ± 6.6	1.17 [1.14, 1.21]	.923	.979	.987

[†]our SMO converged on this cell in 542 ± 63 s.

on the RBF hard margin and $3.2\times$ faster than kernel SMO on the kernel L_2 margin. Distances agree to five digits, support counts within three, test accuracies exactly. Across all five KL2 cells K-ETA returns the same distance and accuracy as kernel SMO with $4\text{--}6\times$ fewer kernel-column evaluations; in wall-clock it is $3\text{--}4\times$ faster where columns are expensive (`gisette`, `w8a`) and $1.5\times$ slower where they are cheap (`ijcnn1` $d=22$, `covtype` $d=54$), because the $O(n)$ NumPy scan per iteration then dominates. The reference hinge-loss SVC ($C=1$, a different objective) takes 1–84 s with accuracies within one point (Appendix C).

The KHM column shows where the geometry turns against the method. On `ijcnn1` both solvers return the identical solution (distance 0.00430, 819 support vectors, accuracy 0.946), but K-ETA needs $1.5\text{--}2.5 \times 10^5$ iterations and 420 s against LIBSVM’s 1.3 s: the feature-space hulls are 0.004 apart in a space of diameter $\approx \sqrt{2}$, so δ/M in Theorem 1 is of order 10^{-3} , and LIBSVM’s second-order working-set selection with heuristic shrinking is far better suited to that conditioning. On `a9a` and `covtype` the hulls are within 10^{-4} and 5×10^{-4} of touching: K-ETA times out at gap 3–10, while LIBSVM’s $C=10^6$ surrogate stops at its KKT tolerance at a point 6–30 $\times$ farther out than K-ETA’s upper bound (or hits its iteration cap on `covtype`), so neither solver certifies anything, and the hard-margin kernel model is the wrong one there anyway (KL2 accuracies 0.79 vs 0.70–0.75). On `w8a` the feature-space hulls intersect (duplicate points with opposite labels); K-ETA certifies this in 101 s and no hard margin exists.

Real data, dense-support regime (Table 2). On the L_2 soft margins of the overlapping datasets both geometric solvers hit the 600 s cap on every cell while LIBLINEAR finishes in 0.4–23 s. At $C=0.1$ ETA’s iterate already has LIBLINEAR’s test accuracy (0.849, 0.918, 0.987) when the cap is reached with the certificate still at 0.01–0.6: the iterate converges long before the certificate closes. At $C \geq 1$ the augmented hulls ($K + I/C$) approach each other, the certified gap at the cap exceeds one and the iterate is not yet usable. Uncapped, ETA does converge to the certified 10^{-3} gap: on `a9a` ($C=0.1$, ten runs) in $2,908 \pm 138$ s and 95,201 iterations, with primal objective 687.07, identical to LIBLINEAR’s to five digits, and on `w8a` in 5,153 s. Figure 1 shows these traces: the linear-rate signature of Theorem 1, but with 60–65% of the points in the support and near-uniform weights, so that $\text{PW}(Z) \sim 1/n$ and the cost is $O(n^2)$ -type. Centroid initialisation and larger blocks did not help (the former increased the iteration count, the latter traded iterations for per-iteration cost).

6 Related work

Distance duality and the TA [1, 2]; MDM [4] and Gilbert’s algorithm; away steps [5] and the pyramidal-width analysis of FW variants [3], whose pairwise step our MDM pair coincides with; the NPA/reduced-hull view of soft margins [6]; block-coordinate FW [7], which blocks over *co-ordinates* of a product domain rather than aggregating pairs on one polytope; safe screening for SVMs [8, 9], which our activation theorem quantifies for the polytope-distance certificate; and the SMO/LIBSVM/LIBLINEAR solvers used as baselines.

ETA on real data (L_2 soft margin, $C = 0.1$): certified gap to the 10^{-3} stopping rule

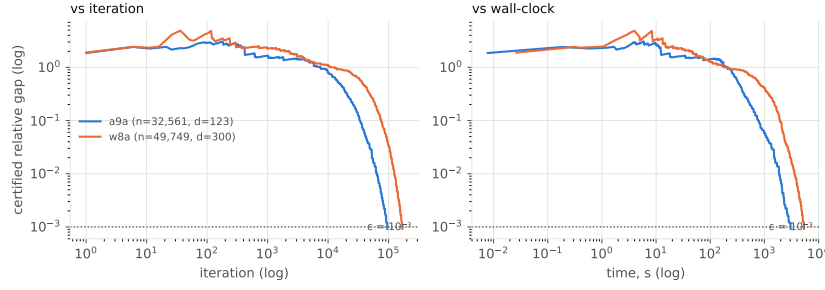


Figure 1: Certified relative gap of ETA on a9a and w8a (L_2 , $C = 0.1$, uncapped), versus iteration and wall-clock: linear-rate convergence in the dense-support regime, at $O(n^2)$ -type cost.

7 Conclusion

Guarded block transfers give the Triangle Algorithm a linear rate with explicit constants and no combinatorial swap-step bounds, together with a screening rule that is both safe and provably effective. The cost is a factor $(k + 1)$ in the exponent, recovered in the near-orthogonal regime where a block realises close to the sum of its k pairwise gains. The experiments locate the boundary the analysis predicts. Where the optimal support is sparse and the hulls are well separated relative to their diameter (high-dimensional hard margins, kernels with expensive columns, linearly separable real data), the solver is competitive with or faster than LIBSVM and kernel SMO and supplies a certificate; where the support is dense or the hulls nearly touch, δ/M is small and primal coordinate descent or second-order SMO is faster by one to two orders of magnitude. Both cases can be recognised during a run from the certified gap and the support size. Limitations: the implementation is NumPy-level (a compiled inner loop would move the practical boundary, not the $O(n^2)$ asymptotics), sparse inputs are not supported, the reduced-hull (ν -SVM) variant is validated but not analysed, and the constants in Theorem 1 are loose.

References

- [1] B. Kalantari. A characterization theorem and an algorithm for a convex hull problem. *Annals of Operations Research*, 226:301–349, 2015.
- [2] M. Gupta and B. Kalantari. An enhanced triangle algorithm for large-scale support vector machine optimization. *JIDMIS*, 3(9s), 2026.
- [3] S. Lacoste-Julien and M. Jaggi. On the global linear convergence of Frank–Wolfe optimization variants. *NeurIPS*, 2015.
- [4] B. F. Mitchell, V. F. Demyanov, and V. N. Malozemov. Finding the point of a polyhedron closest to the origin. *SIAM J. Control*, 12(1):19–26, 1974.
- [5] J. Guélat and P. Marcotte. Some comments on Wolfe’s ‘away step’. *Mathematical Programming*, 35:110–119, 1986.
- [6] K. P. Bennett and E. J. Bredensteiner. Duality and geometry in SVM classifiers. *ICML*, 2000.
- [7] S. Lacoste-Julien, M. Jaggi, M. Schmidt, and P. Pletscher. Block-coordinate Frank–Wolfe optimization for structural SVMs. *ICML*, 2013.
- [8] L. El Ghaoui, V. Viallon, and T. Rabbani. Safe feature elimination in sparse supervised learning. *Pacific J. Optimization*, 8(4):667–698, 2012.
- [9] K. Ogawa, Y. Suzuki, and I. Takeuchi. Safe screening of non-support vectors in pathwise SVM computation. *ICML*, 2013.

215 Statement on the use of AI assistance

216 An AI system (Claude, Anthropic) was used extensively in this work, under the direction of the
 217 authors. It wrote the implementation, drafted the proofs and this manuscript, designed and ran the
 218 experiments, and suggested several of the algorithmic ideas, including the guarded block step, the
 219 away-step fallback and the screening-activation argument. The authors set the research questions,
 220 checked every theorem and proof (a re-derivation during that check corrected a dimensionally in-
 221 consistent constant in an earlier version of Theorem 1), and are responsible for all claims. Each
 222 inequality in the analysis has a numerical test in the released code, every solver is validated against
 223 exact QP solutions, and all experiments can be rerun from the released scripts. Negative results are
 224 reported as obtained.

225 A Proofs

226 **Lemma 4** (Separability). *With the product representation, $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$ where $g_V^{\text{PW}} =$
 227 $\max_i s_i^V - \min_{i \in \text{supp } \alpha} s_i^V \geq 0$ and g_W^{PW} likewise; the same holds for g^{FW} and g^{A} .*

228 *Proof.* $\langle g, v_i - w_j \rangle = s_i^V + s_j^W$ is separable and the support is a product set, so the max over
 229 atoms and the min over the support separate; each side term is nonnegative since the support max
 230 dominates the support min and the global max dominates the support max. $\square$

231 **Lemma 5** (Side transfers are Z -pairwise directions). *For $\alpha_u > 0$ and any $b \in \text{supp } \beta$, $v_r - v_u =$
 232 $(v_r - w_b) - (v_u - w_b)$ is a difference of an atom of Z and an atom in the support, and the transfer
 233 $\gamma \leq \alpha_u$ realises $z \leftarrow z + \gamma(v_r - v_u)$ inside Z .*

234 *Proof of Lemma 1.* Since $\gamma_j \leq \text{gap}_j / \|d_j\|^2$ we have $\gamma_j \text{gap}_j \geq \gamma_j^2 \|d_j\|^2 = b_j^2$, so $S =$
 235 $\sum_j \gamma_j \text{gap}_j \geq \sum_j b_j^2$. By the triangle inequality $\|d\| \leq \sum_j \gamma_j \|d_j\| = \sum_j b_j$ and by Cauchy-
 236 Schwarz $(\sum_j b_j)^2 \leq k' \sum_j b_j^2 \leq k' S$, hence $D \leq k' S$. Exact line search on the 1-smooth F
 237 gives $\Delta_B = S^2 / (2D) \geq S / (2k')$ in the interior case and $\Delta_B = S - D/2 \geq S/2$ at the bound-
 238 ary $t = 1$ (where $D \leq S$). Finally $S \geq \gamma_1 \text{gap}_1 \geq \Delta_{\text{MDM}}$ since all terms are nonnegative and
 239 $\Delta_{\text{MDM}} = \gamma_1 \text{gap}_1 - \gamma_1^2 \|d_1\|^2 / 2$. $\square$

240 *Proof of Lemma 2.* Let $g_S = \max(g_V^{\text{PW}}, g_W^{\text{PW}}) \geq g^{\text{PW}} / 2$. Every direction below is a difference of
 241 two points of one class hull, so its norm is at most $\text{diam}(\text{conv } V) \leq M$. For a feasible direction
 242 with directional derivative $G > 0$, squared norm D and maximal step T , exact line search on
 243 a 1-smooth function gives gain $\geq \min(G^2 / (2D), TG/2)$ (interior optimum, or the clipped step
 244 evaluated with $TD \leq G$). Case (a): the guard's gain is at least the MDM gain, whose derivative is g_S
 245 and whose interior optimum is attained, so gain $\geq g_S^2 / (2M^2) \geq g^{\text{PW}2} / (8M^2)$. Case (b), non-drop:
 246 $g_S = g_{\text{side}}^{\text{FW}} + g_{\text{side}}^{\text{A}}$, so one term is $\geq g_S / 2 \geq g^{\text{PW}} / 4$. If it is the FW term, the toward step (maximal
 247 step 1) gives gain $\geq \min(g_{\text{side}}^{\text{FW}2} / (2M^2), g_{\text{side}}^{\text{FW}} / 2) \geq \min(g^{\text{PW}2} / (32M^2), g^{\text{PW}} / 8)$; if it is the away
 248 term, the away line search is interior (non-drop) and gain $\geq g_{\text{side}}^{\text{A}2} / (2M^2) \geq g^{\text{PW}2} / (32M^2)$. The
 249 algorithm takes the best available candidate. $\square$

250 **Lemma 6** (Geometric strong convexity). *For every $z \in Z$ with a representation whose support
 251 contains an away atom, $g^{\text{PW}}(z) \geq \delta \sqrt{2h(z)}$ and $g^{\text{PW}}(z) \geq g^{\text{FW}}(z) \geq h(z)$.*

252 *Proof.* By the definition of the pyramidal width [3, Def. 3, Thm. 6], for $e = (z^* - z) / \|z^* - z\|$ the
 253 FW atom s and the away atom v satisfy $\langle g, s - v \rangle \geq \delta \langle g, e \rangle$. By 1-strong convexity, $\langle g, z^* - z \rangle \geq$
 254 $h + \frac{1}{2} \|z^* - z\|^2$, so with $r = \|z^* - z\|$, $g^{\text{PW}} \geq \delta(h/r + r/2) \geq \delta \sqrt{2h}$ by AM-GM. Finally
 255 $g^{\text{PW}} = g^{\text{FW}} + g^{\text{A}} \geq g^{\text{FW}} = \max_u \langle g, u - z \rangle \geq \langle g, z^* - z \rangle \geq h$ by convexity. $\square$

256 *Proof of Theorem 1.* By Lemmas 2 and 6 a non-drop iteration gains at least
 257 $\min(2\delta^2 h / (32M^2), h/8) = (\delta/M)^2 h / 16$, using $\delta \leq M$; hence $h_{t+1} \leq (1 - \rho) h_t$. Drop
 258 counting: non-drop iterations add at most k support indices, drop iterations add none and remove at
 259 least one, and the support size stays ≥ 2 from s_0 ; so the number of removals, hence of drops, is at
 260 most $kT_{\text{good}} + s_0$, giving $T_{\text{good}} \geq (T - s_0) / (k + 1)$. All steps, drops included, are descent steps,
 261 so h is monotone. $\square$

Lemma 7 (Two-sided implementation). *If every iteration takes the V-side step and then the W-side step (an iteration counts as a drop if either side drops), every non-drop iteration gains at least $\min(g^{\text{PW}^2}/(128M^2), g^{\text{PW}}/16)$; hence Theorem 1 holds with $\rho_{\text{impl}} = (\delta/M)^2/64$ and $\#\text{drops} \leq 2kT_{\text{good}} + s_0$.*

Proof. Let $G = g^{\text{PW}}(z_t)$. If V carries the larger side gap, the V -step is the analysed step. Otherwise $g_W^{\text{PW}}(z_t) \geq G/2$ and the V -step moves z_t to z' with gain $\Delta_V \geq 0$. Two facts: (i) $\|z' - z_t\|^2 \leq 2\Delta_V$ for an exact line search (interior: equality; boundary $t=1$: $\|z' - z_t\|^2 = D \leq S \leq 2\Delta_V$); (ii) a V -step leaves the W support unchanged and shifts each W score by $\langle z' - z_t, w_j \rangle$, so $g_W^{\text{PW}}(z') \geq g_W^{\text{PW}}(z_t) - \|z' - z_t\| \text{diam}(\text{conv } W) \geq g_W^{\text{PW}}(z_t) - M\|z' - z_t\|$. If $\Delta_V \geq G^2/(128M^2)$ we are done; otherwise $\|z' - z_t\| < G/(8M)$, so $g_W^{\text{PW}}(z') \geq G/2 - G/8 \geq G/4$, and the proof of Lemma 2 with side gap $\geq G/4$ gives the W -step gain $\geq \min(G^2/(128M^2), G/16)$. The rate constant follows as in Theorem 1; each non-drop iteration now adds at most $2k$ indices. $\square$

Proof of Proposition 1. $D = \sum_{i,j} \gamma_i \gamma_j \langle d_i, d_j \rangle \leq \sum_j b_j^2 + \eta \sum_{i \neq j} b_i b_j \leq (1 + \eta(k' - 1)) \sum_j b_j^2 \leq (1 + \eta(k' - 1))S$ by Cauchy–Schwarz on the cross terms; then $\Delta_B = S^2/(2D) \geq S/(2(1 + \eta(k' - 1)))$, and for uncapped pairs $S = \sum_j 2\Delta_j$. $\square$

Proof of Lemma 3. $\text{LB} = \min_{u \in Z} \langle z, u \rangle / \|z\|$ by definition of the supporting hyperplane bound and $\text{UB} = \|z\|$; subtracting, $\text{UB} - \text{LB} = (\langle z, z \rangle - \min_u \langle z, u \rangle) / \|z\| = g^{\text{FW}}/\text{UB}$. At the optimum $0 = \langle z^*, z^* \rangle - \min_u \langle z^*, u \rangle$; subtracting from $g^{\text{FW}}(z), g^{\text{FW}}(z) = (\|z\|^2 - \|z^*\|^2) + (\min_u \langle z^*, u \rangle - \min_u \langle z, u \rangle) \leq 2h + \max_u \langle z^* - z, u \rangle \leq 2h + R\|z - z^*\| \leq 2h + R\sqrt{2h}$ by 1-strong convexity. $\square$

Proof of Theorem 2. Strong convexity gives $\|z - z^*\| \leq r$ for the screening radius $r = \sqrt{\text{UB}^2 - \max(\text{LB}, 0)^2}$. Let $m^* = \min_j \langle z^*, v_j \rangle$ and pick $v' \in F_V$; since $v_{\min}$ minimises $\langle z, \cdot \rangle$, $\langle z^*, v_{\min} \rangle - m^* = \langle z, v_{\min} - v' \rangle + \langle z^* - z, v_{\min} - v' \rangle \leq rD$. Hence $\langle z, v_i - v_{\min} \rangle = \langle z^*, v_i - v_{\min} \rangle + \langle z - z^*, v_i - v_{\min} \rangle \geq \tau_i - rD - r\|v_i - v_{\min}\|$, and the rule fires when this exceeds $r\|v_i - v_{\min}\|$, for which $\tau_i \geq 3rD$ suffices. For the activation time, $r^2 = (\text{UB} - \text{LB})(\text{UB} + \text{LB}) \leq 2R(\text{UB} - \text{LB}) = 2Rg^{\text{FW}}/\text{UB} \leq (2R/\delta^*)(2h + R\sqrt{2h}) \leq (4R^2/\delta^*)\sqrt{2h}$ for $h \leq R^2/2$ (Lemma 3); requiring $r^2 \leq \tau^2/(9D^2)$ yields the stated condition on h_t , and Theorem 1 with $\log(1/(1 - \rho)) \geq \rho$ converts it into the iteration bound. Permanence follows from the safety of the rule. $\square$

290 B Exact reductions

L_2 soft margin. The problem $\min \frac{1}{2}\|w\|^2 + \frac{C}{2} \sum_i \xi_i^2$ is the hard-margin problem on the augmented points $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$, whose Gram matrix is $K + I/C$ and whose augmented hulls never intersect. The augmentation is never materialised: the extra coordinates of an iterate are exactly its convex weights, so every augmented inner product is a cached original-space product plus a sparse diagonal correction, $\langle \tilde{p}, \tilde{v}_i \rangle = \langle p, v_i \rangle + \alpha_i/C$ and $\|\tilde{p}\|^2 = \|p\|^2 + \|\alpha\|^2/C$. The separator is recovered as $w = 2(p - q)/\delta_C^2$, $b = (\|\tilde{q}\|^2 - \|\tilde{p}\|^2)/\delta_C^2$, and the primal value is $P = 2/\delta_C^2$. Validation: the solver agrees with an exact QP on the explicitly augmented problem to 10^{-10} across 298 random instances and values of C ; the regularisation path δ_C is monotone in C ; on a9a ($C=0.1$) its certified primal 687.07 equals LIBLINEAR's to five digits.

ν -SVM. The L_1 (hinge) soft margin is the nearest-point problem between *reduced* convex hulls $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$, $\mu = 2/(\nu\ell)$ [6]. Two changes: the extreme-point oracle becomes a capped top- k blend (weight μ on the $\lfloor 1/\mu \rfloor$ best points in the search direction, via a partial sort, still $O(n)$), and the primary step is the capped MDM transfer with blended toward steps as fallback; caching and the reduced support-function lower bound carry over unchanged. Validation: agreement with the exact box-constrained QP to 10^{-9} – 10^{-14} and with LIBSVM's NuSVC to $\sim 10^{-10}$ through the mapping $\delta = \|w_{\text{svc}}\|/S$, S the per-class sum of the rescaled dual coefficients. The reduced hull is a polytope, so (1) still applies, but its vertices are blends and the capped transfer is not the pairwise step analysed in §4; the analysis of this variant is left open.

Kernels. The algorithm touches data only through inner products, so kernelisation changes plumbing, not algorithm: Gram columns become kernel columns $k(V, x_i)$, squared norms become kernel diagonals, iterates exist purely as convex weights (the exact cache refresh is reconstructed from the support’s cached kernel columns), and the classifier is the kernel expansion of the witness-pair bi-sector. A ridge $K + I/C$ gives the kernel L_2 margin by the same diagonal trick. Validation: the linear kernel reproduces the Euclidean solver to 10^{-12} ; RBF distances match the exact kernel QP to 10^{-9} and LIBSVM’s hard-margin solution, via $\|w_{\mathcal{H}}\|^2 = (\alpha \circ y)^\top K(\alpha \circ y)$, to 10^{-5} ; kernel SMO matches the QP to 10^{-8} .

317 C Experimental details

Solver settings and tolerance equivalence. ETA: block transfers with $k = 16$, guarded, away fallback, pairwise anti-zig-zag, stop at certified relative gap 10^{-3} , iteration cap 2×10^6 , 600 s cap; screening (every 25 iterations) only in the LIN cell, where it never activated at 10^{-3} , as Theorem 2 predicts. Our SMO: maximal-violating-pair selection, KKT gap 10^{-3} , same caps; hard margin as $C = 10^{12}$; kernel SMO on $K + I/C$. LIBLINEAR (scikit-learn 1.8 LinearSVC): squared hinge with parameter $C/2$, so that its objective equals $\frac{1}{2}\|w\|^2 + \frac{C}{2} \sum \xi_i^2$; tolerance 10^{-6} , chosen so that its final primal matches ETA’s certified value to five digits on every cell where ETA converges (a9a 687.07 vs 687.07; ijcn1 550.6 vs 550.5 at ETA gap 0.01); intercept scaling 100; iteration cap 2×10^5 . LIBSVM (scikit-learn SVC): tolerance 10^{-3} , 1000 MB cache, iteration cap 2×10^6 (reported as *maxiter* when hit); hard margins as $C = 10^6$; RBF with $\gamma = 1/d$. Oracle counts: ETA counts Gram or kernel columns evaluated against a full class ($O(nd)$ or $O(n_k d)$ each); SMO counts kernel rows, the same unit.

Data and seeds. a9a, w8a, ijcn1 and gisette use the LIBSVM training and test files; covtype (binary, scaled) is subsampled to 100,000 points and split 75/25 by seed. Features are standardised on the training set. Kernel cells use a class-balanced subsample of at most 5,000 points per class (w8a’s minority class has 1,479 points, hence $n_k = 6,479$; gisette uses all 6,000). Seeds vary the covtype split, the kernel subsamples and the timing environment; on datasets with fixed splits the deterministic solvers return identical solutions across seeds, which is why several accuracy CIs are zero. Confidence intervals are t -based 95% intervals over seeds.

Hardware and concurrency. One dedicated 32-vCPU Intel Xeon 2.6 GHz Lightning Studio, 122 GB RAM, Linux, Python 3.12, NumPy 2.4 with OpenBLAS 0.3.31, OMP_NUM_THREADS=1; the L_2 battery ran 14 cells and the regime battery 12 cells concurrently. The uncapped a9a/w8a runs in Figure 1 were made earlier on the same machine under a different load (LIBLINEAR took 12 and 48 s there against 4 and 23 s in the capped battery), so ratios across the two runs carry a factor of roughly two of uncertainty; ratios within a battery do not.

Reference solver in the KL2 cells. The hinge-loss SVC(rbf, C=1), a different objective, took 8.6 ± 2.9 (a9a), 3.4 ± 0.3 (covtype), 83.5 ± 13.1 (gisette), 1.1 ± 0.0 (ijcn1) and 4.9 ± 1.2 s (w8a), with test accuracies 0.789, 0.783, 0.977, 0.950 and 0.975, within one point of the L_2 solutions and on either side of them.

Step ablation, blocks and shrinking (synthetic). Iteration counts at $\varepsilon = 10^{-6}$ (single instances): $d=30, n=200$: toward 158,941, midpoint 148,996, away 106, pairwise 181; $d=100, n=1,000$: toward and midpoint fail within 300,000, away 15,886, pairwise 3,921. At the working tolerance 10^{-3} the pairwise/away gain over the midpoint heuristic is only $\sim 11\%$, which is why the heuristic sufficed in [2]. Block transfers cut iterations $4\text{--}30\times$ relative to single MDM (e.g. $43,486 \rightarrow 1,551$ on a dense-support L_2 instance at $d=1,000$) and are the only configuration that reaches the certified tolerance on MNIST odd-vs-even. Figure 2 shows the screening speed-ups and Figure 3 the consolidated comparison: hard margins 0.24 vs LIBSVM’s 0.39 s at $d=1,000$ and 8.6 vs 20.5 s at $d=10,000$; 0.21 vs 0.42 s at $\varepsilon=10^{-5}$ (where the paper-style configuration fails to converge); RBF hard margin 0.76 vs 1.28 s; ν -SVM 18.5 vs 21.9 s; synthetic L_2 ($d=1,000, C=1$) 8.9 vs LIBLINEAR’s 13.8 s; MNIST odd-vs-even 82.8 vs LIBLINEAR’s 55.0 s. Distances agree within tolerance on every row.

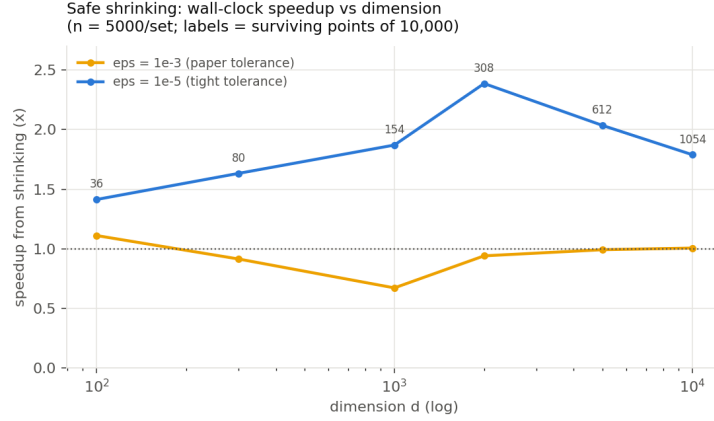


Figure 2: Safe screening: wall-clock speed-up vs dimension at two tolerances on the separated synthetic protocol; labels are surviving points of 10,000.

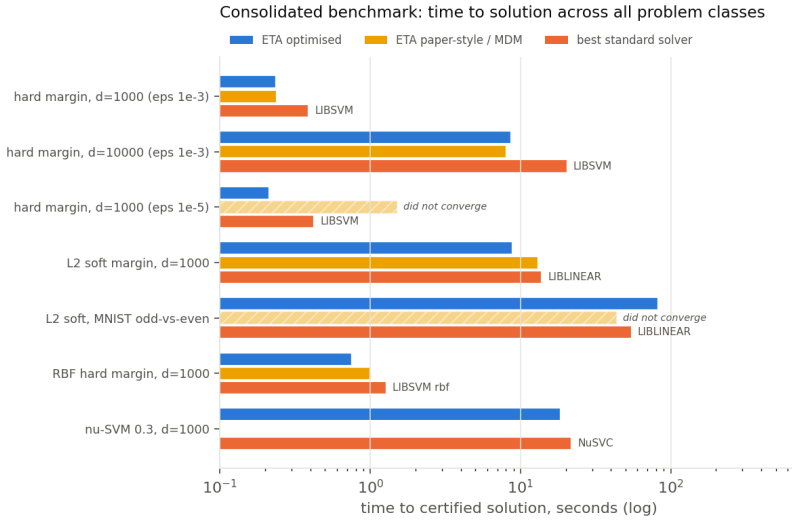


Figure 3: Time to certified solution on seven problem classes: optimised ETA vs the paper-style configuration vs the best standard solver per class.

D Numerical verification of the analysis

Lemma 1 was checked on 4,000 random ensembles including heavily clipped and boundary cases (worst slack at machine precision); Proposition 1 on 500 near-orthogonal ensembles; Lemma 3 to 10^{-9} at arbitrary iterates; the three ingredients of Lemma 7 ($\|z' - z_t\|^2 \leq 2\Delta_V$, the $M\|z' - z_t\|$ Lipschitz bound on the side gap, and the resulting per-iteration gain) at machine precision; and Theorem 2 on instances whose optimal face was computed by an exact QP: at every activating round all zero-weight non-face points were screened, and the final working set equalled the optimal face exactly. Over 5,000 instrumented live block steps none increased the objective. All checks are test files in the released code.